Attention-Based Segmentation from Stable Diffusion 1.5

Ablations, interpretation, and cross-dataset generalization

October 2026

1 Experimental protocol

Each released prompt and seed is regenerated for 30 inference steps. Maps are scored against the released mask and then discarded. mIoU is the 20-class macro average of pooled foreground IoU. OVAM is reproduced with its released dataset-specific thresholds and DenseCRF; token optimization is excluded.

Our revised final configuration retains all 16 CA layers. Frozen CA head filtering is applied to layers 0, 2, 5, 6, 7, 9, 10, 11, 12, 13, 14, and 15; layers 1, 3, 4, and 8 use their all-head means. CA is aggregated over calls 0–30. The final 64×64 decoder SA module (U3.2) is applied once using selected SA heads $\{0, 1, 2, 4, 5, 6\}$ and calls 15–30. Cross-attention attribution follows the OVAM-compatible single-token convention. The output uses percentile 1–99 normalization, threshold 0.45, and removal of connected components below 1% of the image.

2 Cross-attention ablations

2.1 Denoising time

Fixed configuration: CA only on all 490 VOC-Sim images; all 15 non-middle CA modules at 16, 32, and 64 resolution; all eight heads and all class-token pieces; conditional CFG branch; OVAM max-after-resize normalization; threshold 0.40; no SA fusion, DenseCRF, or component cleanup. Only the included denoising calls change.

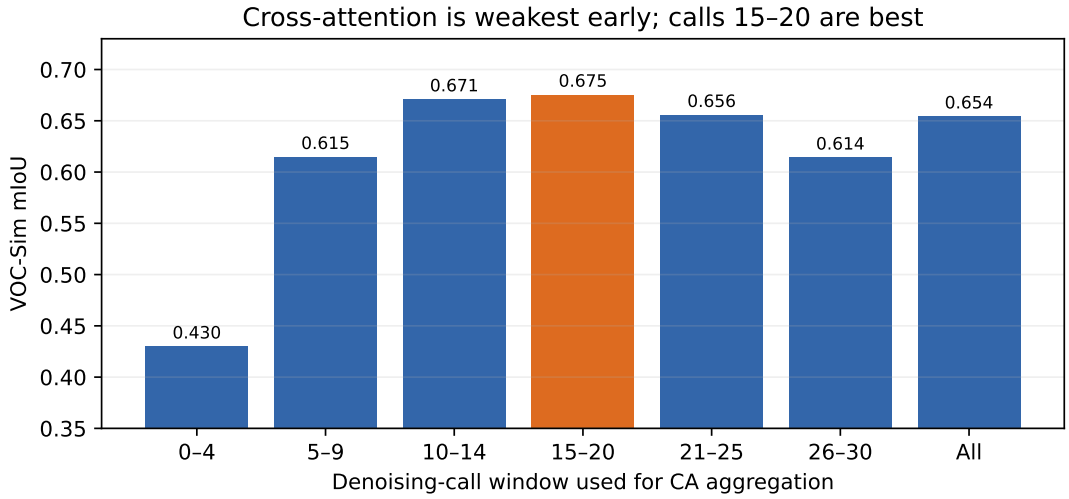


Figure 1: CA-only time-window ablation on all 490 VOC-Sim images. Very early calls are poorly localized; calls 15–20 are strongest in this aggregation.

CA calls	0–4	5–9	10–14	15–20	21–25	26–30	All
mIoU	0.430	0.615	0.671	**0.675**	0.656	0.614	0.654

Table 1: CA time-window mIoU under the fixed configuration above.

2.2 Resolution

Fixed configuration: CA only on all 490 VOC-Sim images; all denoising calls 0–30; every CA module at the listed resolutions; all eight heads and all class-token pieces; conditional CFG branch; OVAM max-after-resize normalization; threshold 0.40; no SA fusion, DenseCRF, or component cleanup. Only the retained resolutions change.

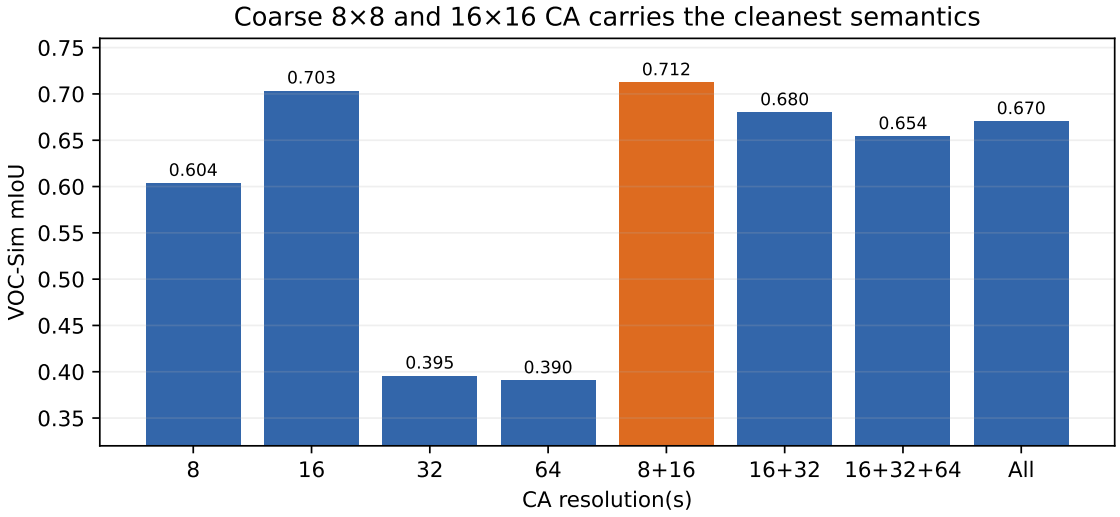


Figure 2: CA-only resolution screening. Combining 8×8 and 16×16 maps is best in isolation. The end-to-end revised system retains all layers and therefore all four resolutions.

CA resolution	8	16	32	64	8+16	16+32	16+32+64	All
mIoU	0.604	0.703	0.395	0.390	**0.712**	0.680	0.654	0.670

Table 2: CA resolution mIoU under the fixed configuration above.

Coarse 8×8 and 16×16 CA carries cleaner semantics in isolation. The transfer experiment shows that discarding the other resolutions together with their layers does not improve the final system.

2.3 Individual layers

Fixed configuration: CA only on all 490 VOC-Sim images; one listed module at a time; all denoising calls 0–30; all eight heads and all class-token pieces; conditional CFG branch; percentile 1–99 normalization and threshold 0.45; no SA fusion or component cleanup. This diagnostic averages image IoU, so its values should be used to rank modules rather than compared numerically with the class-macro mIoU tables.

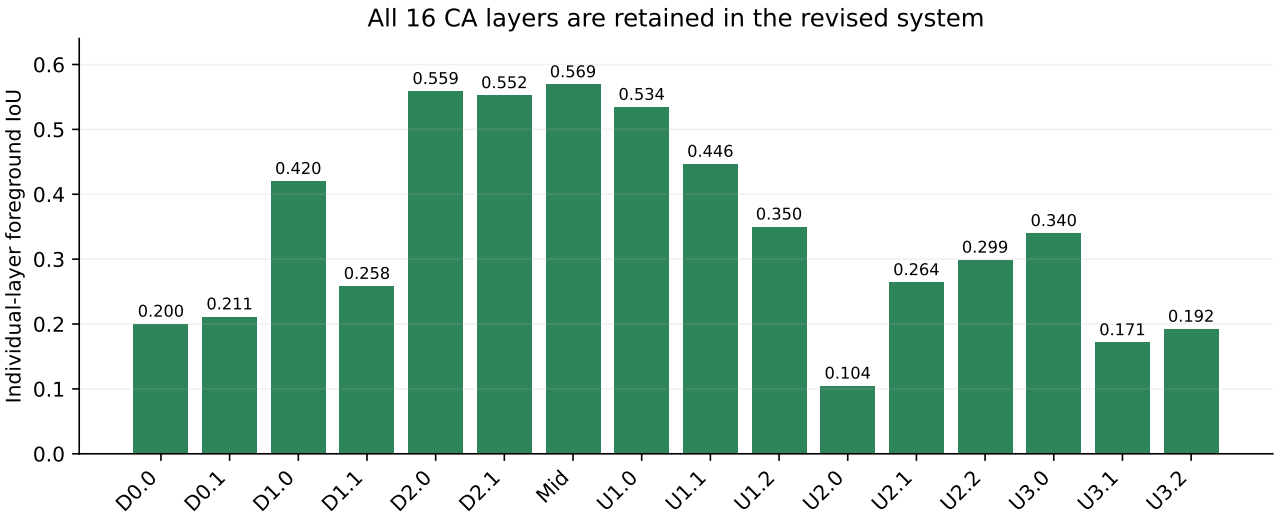


Figure 3: Mean image IoU for individual CA modules. Individual quality varies substantially, with the middle 8×8 block strongest.

Layer	Res.	Mean image IoU	Layer	Res.	Mean image IoU
D0.0	64	0.200	U1.1	16	0.446
D0.1	64	0.211	U1.2	16	0.350
D1.0	32	0.420	U2.0	32	0.104
D1.1	32	0.258	U2.1	32	0.264
D2.0	16	0.559	U2.2	32	0.299
D2.1	16	0.552	U3.0	64	0.340
Mid	8	**0.569**	U3.1	64	0.171
U1.0	16	0.534	U3.2	64	0.192

Table 3: Individual CA-module screening; every listed module is retained by the final system.

The middle 8×8 block is strongest individually. Individual rank is not sufficient for layer removal: the earlier filtering removed both strong and weak layers and reduced transfer performance.

2.4 What CA head filtering contributes

Fixed configuration: all 16 CA layers and CA calls 0–30. The filtered seed uses frozen top-four head rankings for layers $\{0, 2, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15\}$ and all-head means for layers $\{1, 3, 4, 8\}$; the control uses all eight heads everywhere. Both use percentile 1–99 normalization. "After matched SA" uses one pass through U3.2 at 64×64, SA calls 15–30, SA heads $\{0, 1, 2, 4, 5, 6\}$, threshold 0.45, and 1% component cleanup.

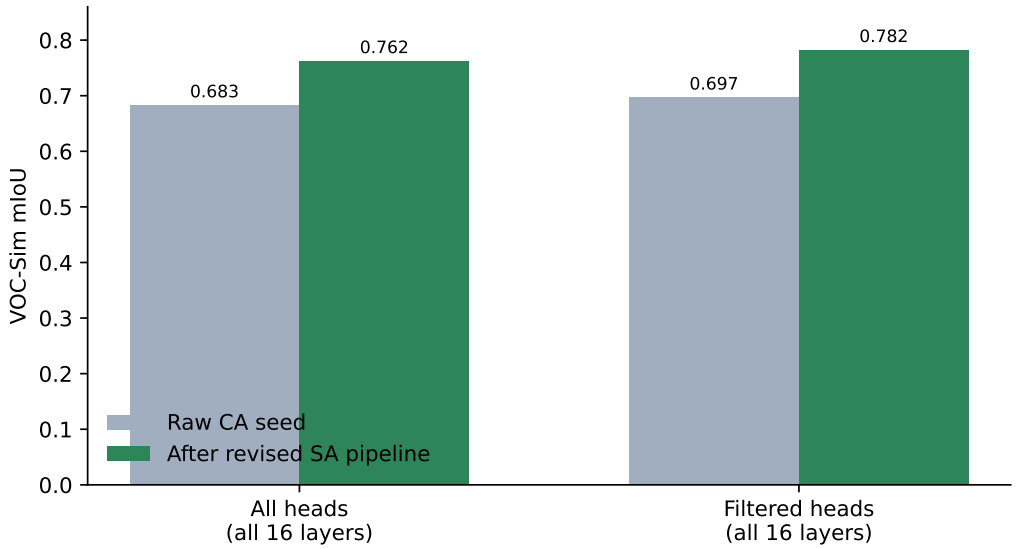


Figure 4: Both seeds retain all 16 CA layers. Head filtering improves the raw seed and its output after the revised one-pass SA pipeline.

CA seed	Raw mIoU	After matched SA	SA gain
All 16 layers/all heads	0.6830	0.7625	+0.0795
All 16 layers/filtered heads	**0.6969**	**0.7824**	+0.0855

Table 4: Matched CA-head comparison for the revised all-layer system.

Restoring all layers improves cross-dataset transfer. Within that all-layer system, CA head filtering contributes 0.0199 mIoU on VOC-Sim and 0.0069 on COCO-Cap after the complete pipeline.

3 Self-attention ablations

3.1 Which SA module?

Fixed configuration: all 490 VOC-Sim images; a CA seed formed from the frozen top-four heads of CA layers $\{0, 2, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15\}$, aggregated over CA calls 0–30 and normalized by percentiles 1–99; all eight SA heads; SA calls 15–30; four propagation passes; percentile 1–99 output normalization; threshold 0.45; no component cleanup. Only the SA module changes. As in the layer diagnostic, the reported score is mean image IoU.

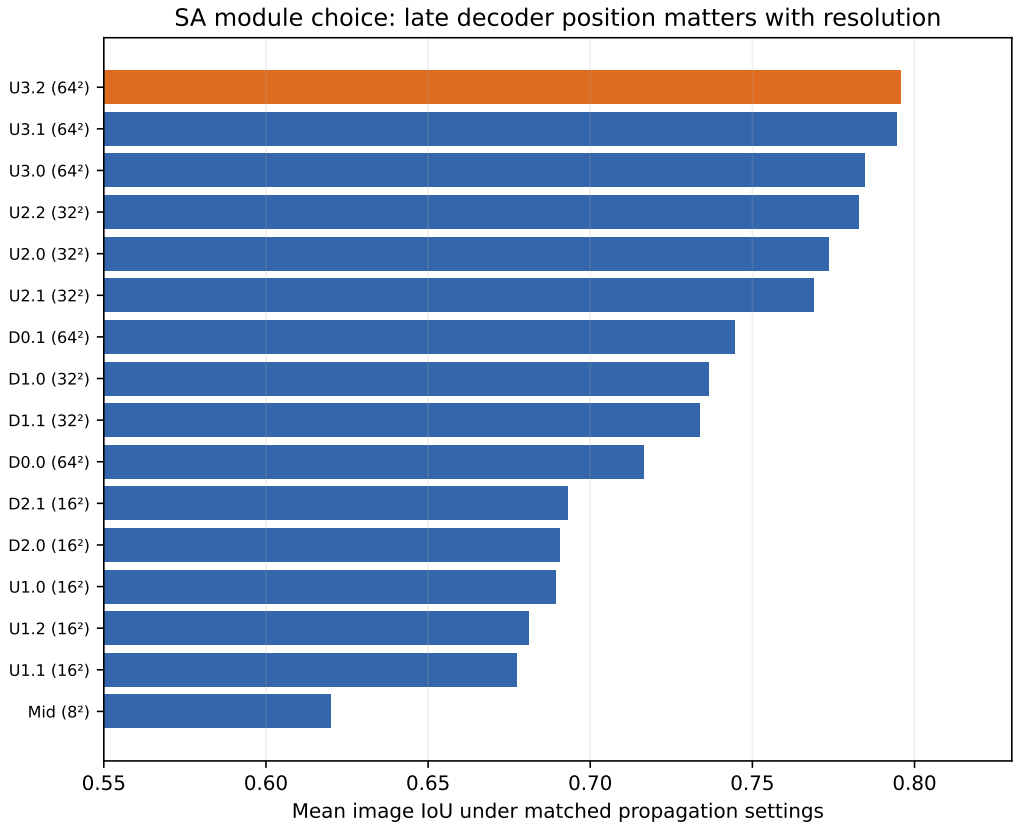


Figure 5: Mean image IoU for every SA module under the fixed configuration above. The final 64×64 upsampling module is best.

SA module	Res.	Mean image IoU	SA module	Res.	Mean image IoU
D0.0	64	0.717	U1.1	16	0.677
D0.1	64	0.745	U1.2	16	0.681
D1.0	32	0.737	U2.0	32	0.773
D1.1	32	0.734	U2.1	32	0.769
D2.0	16	0.691	U2.2	32	0.783
D2.1	16	0.693	U3.0	64	0.785
Mid	8	0.620	U3.1	64	0.794
U1.0	16	0.689	U3.2	64	**0.796**

Table 5: SA-module screening under matched settings.

The last 64×64 decoder module is best. Both decoder position and resolution matter; the early 64×64 modules are much weaker.

3.2 Which denoising calls?

Fixed configuration: all 16 CA layers and CA calls 0–30; frozen CA head filtering; one pass through U3.2 at 64×64; SA heads $\{0, 1, 2, 4, 5, 6\}$; percentile 1–99 normalization; threshold 0.45; and 1% component cleanup. Only the SA call set changes. The same frozen configuration is evaluated on all 490 VOC-Sim and 707 COCO-Cap images.

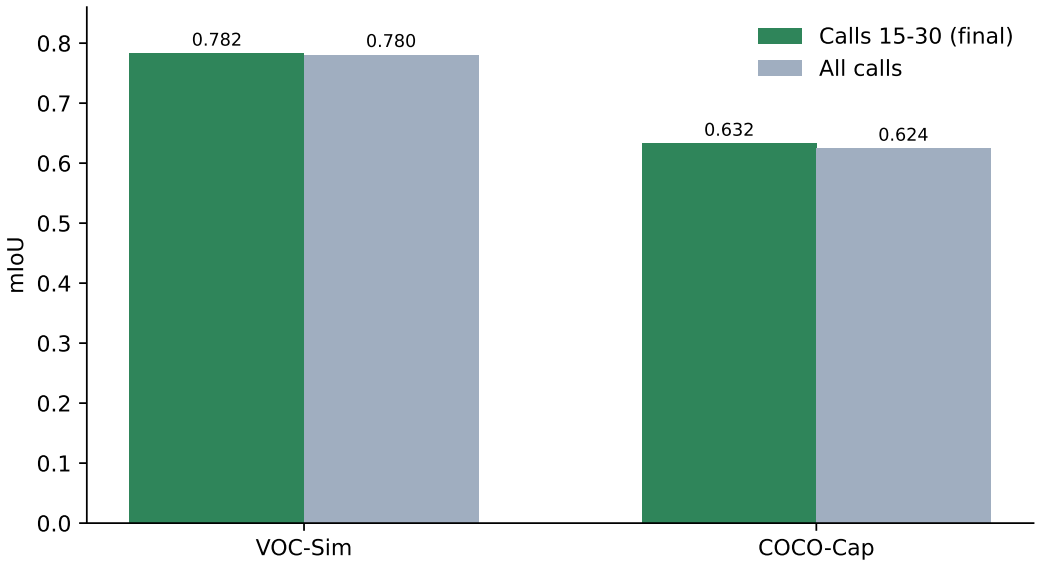


Figure 6: Matched timestep-window screening in the selected-SA-head predecessor. Calls 15–30 improve both datasets.

Dataset	Calls 15–30	All calls	Late-window gain
VOC-Sim	**0.7824**	0.7802	+0.0022
COCO-Cap	**0.6322**	0.6240	+0.0083

Table 6: Leave-one-change comparison for SA timestep filtering.

Late calls provide a small VOC improvement and a clearer COCO-Cap improvement. The final system therefore retains calls 15–30.

3.3 Which SA heads?

Fixed configuration: all 16 CA layers and CA calls 0–30; frozen CA head filtering; one pass through U3.2 at 64×64; SA calls 15–30; percentile 1–99 normalization; threshold 0.45; and 1% component cleanup. Only the SA head set changes. "Selected" means $\{0, 1, 2, 4, 5, 6\}$; "all" means heads 0–7.

Dataset	Selected heads	All heads	Selected-head gain
VOC-Sim	0.7824	**0.7837**	-0.0013
COCO-Cap	**0.6322**	0.6274	+0.0049

Table 7: Leave-one-change comparison of the SA-head sets on both datasets.

All SA heads improve VOC-Sim by 0.0013 mIoU, but selected heads improve the untuned COCO-Cap result by 0.0049. The final system retains the selected set.

4 What changes in the final mask?

Measure	Implemented OVAM	Final method	Change
Foreground precision	0.710	0.866	+0.156
Foreground recall	0.895	0.876	−0.019
Predicted foreground area	0.467	0.375	−0.092
Ground-truth foreground area	0.370	0.370	—

Table 8: Binary-mask diagnostics for the revised final system over 490 VOC-Sim images.

Precision rises from 0.710 to 0.866 while recall changes from 0.895 to 0.876. The revised mask nearly matches the true foreground area, showing that the gain mainly comes from suppressing false foreground.

Class-area quartile	Mean gain	Classes
Q1 (smallest)	+0.0969	bird, aeroplane, bottle, chair, boat
Q2	+0.0919	table, train, bicycle, car, bus
Q3	+0.0747	motorbike, horse, dog, cow, person
Q4 (largest)	+0.0662	sheep, sofa, plant, tv, cat

Table 9: Revised-system gain grouped by class-average foreground area.

Every class-size quartile improves. The declining trend indicates that OVAM already handles many large foreground objects well, while the revised system has more room to suppress false foreground around smaller objects.

5 Generalization to COCO-Cap

The final configuration uses all 16 CA layers, CA calls 0–30, frozen CA head filtering, one U3.2 SA pass, SA calls 15–30, and selected SA heads $\{0, 1, 2, 4, 5, 6\}$.

Dataset	Method	Images	Classes	mIoU	Gain over OVAM
VOC-Sim	Implemented OVAM	490	20	0.6934	—
VOC-Sim	Final method	490	20	**0.7824**	**+0.0890**
COCO-Cap	OVAM	707	20	0.5719	—
COCO-Cap	Final method, frozen	707	20	**0.6322**	**+0.0603**

Table 10: Primary comparison. VOC-Sim is the development dataset; COCO-Cap is the untuned generalization dataset.

The authors' released VOC-Sim masks score 0.7036. We use our live 0.6934 implementation for the matched VOC-Sim comparison because both methods use the same regenerated trajectories.